

DIGITAL PRODUCT DESIGN

Lecture 3: EVC & Market Analysis

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***Based on slides provided by Catherine Tucker**

Outline

- Ideation Review
- Economic Value to the Customer
- Competitive Analysis
- EVC Calculations for Projects

ECONOMIC VALUE TO THE CUSTOMER

Economic Value to the Customer

- The **economic value to the customer** is the value created for the customer assuming they are fully informed about the product and competitors
- Key idea: customer will purchase our product if the net value of our product exceeds net value of the competitors

$$V_o - P_o \geq V_c - P_c$$

- V_o : Value of our own product
- P_o : Price of our own product
- V_c : Value of competitor c
- P_c : Price of competitor c

Economic Value to the Customer, Part 2

$$V_o - P_o \geq V_c - P_c$$

Economic Value to the Customer, Part 3

- Key idea: customer will purchase our product if value of our product exceeds value of the competitors

$$V_o - P_o \geq V_c - P_c$$

- Rearranging:

$$P_o \leq [V_o - V_c] + P_c$$

- $[V_o - V_c]$: Differentiation value of our product
- Willing to pay up to:

$$\text{Economic Value}_o = [V_o - V_c] + P_c$$

Economic Value to the Customer, Part 4

- Useful way to reason about the value our product creates for customers
 - How does this solve a problem better than competitor?
 - How does our competitors price impact the value of our offering?
- EVC summarizes the value created by our offering
 - Corresponds to a theoretical upper bound of how much customer will pay

A Concrete Example – Hydroponic Tulips

- New container named TulipBowl allows tulip farmers to grow tulips hydroponically
 - Use water rather than soil to grow bulbs
- Current tulip growing containers retail at \$10 each
- New container:
 - Increases yield (reduces waste)
 - Uses cheaper mineral nutrients vs pesticides



A Concrete Example – Hydroponic Tulips, Part 2

Feature	Existing Container	New Container	Diff Value
Pesticides	\$2.50	0	\$2.50
Mineral Solution	0	\$1.50	-\$1.50
Reduces Disease	\$80 if happens Pr(Disease)=0.1	0	\$8
Total Diff Value			\$9

- Total differentiation value: \$9
- Cost of alternative: \$10

$$EVC = \$9 + \$10 = \$19$$

Step-by-Step Guide

1. Identify the end benefit
2. Identify the closest competitor
3. Identify closest competitor's price
4. Measure differentiation values
5. Calculate EVC
6. Repeat for various configurations and customer segments

1. Identify the End Benefit

- Hopefully we have a clear value proposition for our new product or feature
- Ideation process is customer and solution oriented
- Should have a clear thesis on how our product delivers value to end user

2. Identify the Competitors

- Sleight of hand in the definition of EVC

$$V_o - P_o \geq \max_c V_c - P_c$$

- Customers will only purchase if value of our product exceeds value of all competitors
 - “Closest” competitor is the competitor that maximizes the right hand side
- Good to identify many competitors
 - You can do this calculation for many competitors or alternatives
 - Important to be honest about competitors
 - Doing nothing is often the closest competitor (and doing nothing is free)
 - Risk of using aspirational closest competitor

2. Identify the Competitors



3. Identify the Competitor Prices



4. Measure Differentiation Values

- If product is developed, can do some field tests to assess value (e.g. A/B test)
- Use the results of survey analysis (e.g. conjoint feature values)
 - This gives us a dollar value for various features
 - Make sure survey respondents are representative of customer segment!
- Also easy to do when your product saves the customer costs
 - Example: Automation software, medical devices, reducing waste

4. Measure Differentiation Values

Announcing Cresta Voice Virtual Agent: What Sets It Apart & What the Market Is Getting Wrong - Read the post

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AI generated answer

Eligible plans that include drug coverage in Crestview, California for ages 65+ are listed from least to most expensive:

- MediPlus RX
- Guardian Complete
- Guardian Complete +

Source: 65+ Plans in Crestview County CA

Was this helpful?

Q What are the eligible plans for the customer?

Conversation reasons

100.0K

• Upgrade Service	15.0%	15,000
• New Customer	10.0%	10,000
• Cancel Account	10.0%	10,000
• Move Transfer	10.0%	10,000
• Phone Request	10.0%	10,000

Calculate EVC for Various Segments

- Once we have differentiation value and competitors price, easy to calculate EVC
- EVC can vary widely by customer segment
 - Differentiation likely differs by segment
 - Closest competitor can differ for various segments

Uses of EVC

- EVC helps us understand the potential of a market
 - Can be done with very little data
 - As you collect more data, can refine estimates of differentiation value
- Feature prioritization
 - What features differentiate our product from competitors?
- Also frequently used in pricing
 - Gives a sense of the maximum you **can** charge
 - Best practice to discount off of EVC
 - Gives sales an answer to the question “How can you charge so much?!?”

COMPETITIVE ANALYSIS AND GAME THEORY

EVC Calculations Have a Strong Assumption

- Competitors respond to changes to the competitive landscape
- How can competitors respond?
 - Reduce prices
 - Imitate product
 - Invest in new features
 - ...
- Game theory provides the tools to study strategic interactions between players
 - Players here are different firms (including us!)

Game Theory Key Ideas

- Individuals acting strategically in their best interests taking into account other player's strategies
 - If we change the game (i.e. market) by introducing a new feature / product, how will competitors respond?
- Nash Equilibrium: No player can profit from a deviation given the strategies of others
- Game theory is an incredibly rich field
 - Not possible to cover in a 6 week class, let alone half a class!
 - Without formal modeling, we can reason about equilibrium concepts
 - i.e. if we undercut our competitors' price, how will they respond?

Finding Equilibria

- General idea: Calculate best-response functions of each player
 - This is function returns the best move you can make in response to the other players' strategies
 - Equilibrium is when every player is best-responding to everyone else

	Cooperate	Defect
Cooperate	-1, -1	-10, 0
Defect	0, -10	-5, -5

The prisoner's dilemma

	Swerve	Straight
Swerve	Tie, Tie	Lose, Win
Straight	Win, Lose	Crash, Crash

Fig. 1: A [payoff matrix](#) of Chicken

Game Theory to Understand Competitive Response

- If we enter the market with a new product or feature, how will competitor respond?
- We can reason about their best-response to our new product / feature
 - Competitors notice a negative EVC of their product
- Best response likely includes a combination of:
 - Price decreases to make competitor's EVC calculations look more favorable
 - Imitation / investment to reduce the disparity in differentiation value

Game Theory to Understand Competitive Response, Part 2

- Important question at design phase: Is competitive advantage sustainable?
- How will our EVC calculations change after the competitor responds by cutting prices or changing their product offering?
- Two scenarios:
 - Large differentiation value: prevent competition from copying innovation?
 - Patents, proprietary technology, proprietary data, network effects, switching costs, ...
 - Small differentiation value:
 - Price war has at most two winners: Customer and low-cost producer

PRACTICE EVC

Practice EVC

- Let's do an EVC analysis for the product you are using in your project
- Gather as a team, work through the various steps for the product

Step-by-Step Guide

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SUMMARY

Recap

- EVC helps us understand the potential of a market
 - Theoretical maximum willingness to pay for our product
- Feature prioritization
 - What features differentiate our product from competitors?
- Frequently used in pricing
- Important to reason about how competitors will respond to our offering

Project

■ Key Dates

- 10/31: Team formed and product selected (email me your team)
 - Also self-enroll for groups on Brightspace!
- 11/6: Midpoint check-ins
- 12/9-12/11: Final project presentations + final report due